Appendix I. Burkina Faso: Joint Bank/Fund Debt Sustainability Analysis

Risks to the debt sustainability outlook have increased since the previous debt sustainability analysis (DSA) update in December 2007. *Following the reclassification of Burkina Faso as a medium performer based on the World Bank's Country Policy and Institutional Assessment (CPIA) rating, one indicative threshold (the NPV of debt-to-exports ratio) is breached in the baseline scenario. This breach leads to a classification of high risk of debt distress.*

I. External debt sustainability analysis[1]

A. What Has Changed Since the Last DSA?[2]

1. Following a weaker rating in the World Bank's CPIA rating for 2006 and 2007, Burkina Faso is now classified as a medium performer. In the joint DSA from April 2007, Burkina Faso was a strong performer based on the 2005 CPIA rating, and this classification was maintained for the DSA update by Fund staff in December 2007. However, the average CPIA rating for Burkina Faso for the period 2005-2007 has declined below the indicative threshold for strong performers.[3] Consequently Burkina Faso has now to be reclassified as a medium performer. The main causes for the decline in the CPIA ratings are a drop in 2006 of both the debt policy rating and the transparency and accountability rating, by half a point each.

2. **The reclassification to medium performer implies lower indicative thresholds for debt indicators in the DSF**. This is particularly relevant for the NPV of debt-to-exports ratio, for which the baseline projection now breaches the indicative threshold starting in 2013. For other debt indicators, baseline projections and some stress tests remain well below their indicative thresholds.

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[1] This DSA has been prepared jointly by staffs of the World Bank and Fund. The African Development Bank participated in the DSA mission.

[2] For the joint DSA from April 2007, see IMF Country Report No. 07/153; for the DSA update from December 2007 prepared by Fund staff, see IMF Country Report No. 08/168. The comparison to changes since the last DSA refers to the December 2007 update, unless otherwise indicated.

[3] Results for the 2006 CPIA rating became available in summer 2006; the 2006 rating was 3.69, below the threshold of 3.75 for strong performers. Results for the 2007 CPIA rating, as well as the 3-year average over 2005–07, also point to a medium policy performance.

3. **The medium-term export outlook has been revised upwards (Figure I.1, Panel 1) relative to projections in the DSA update from December 2007, mostly because of two factors**:[4]

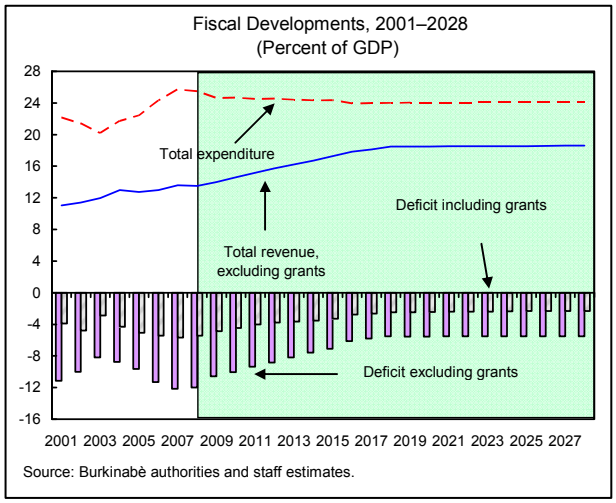
- **Higher commodity prices**: Gold prices have increased by about 35 percent in US Dollars and about 25 percent in euro-terms; this has a significant effect on export projections because of the start-up of large commercial gold mining operations beginning in 2008.
- **Reassessment of medium-term export potential**: the most significant change is an upward revision for medium-term cotton exports, based on sector-specific World Bank input, which reflects a rebound to pre-crisis level in cotton output followed by gradual productivity improvements due to the introduction of higher-yield GM cotton seeds, better storage facilities, and advanced agricultural techniques (Figure I.1, Panel 2). Due to the constrained availability of arable lands, the share of cotton exports in percent of GDP is projected to decline markedly over the long-term (Figure I.1, Panel 3). Compared to the relatively stable export-to-GDP ratio in the previous DSA, the overall export-to-GDP profile is now hump-shaped. The peak in 2015 arises because of a strong expansion in the mining sector and a rebound in the cotton sector; in the long-run, the role of both sectors declines in percent of GDP, while other sectors, like services, livestock and light manufacturing, gradually increase in significance.

[4] The definition of service exports has been aligned with conventions in the Balance of Payments Manual, Fifth Edition; in contrast to the previous definition, the Fifth Edition treats the income account (formerly classified as factor services) separately from the service account. As a result, exports of services are now somewhat smaller than had been previously the case. This accounts for the downward revision of historical data shown in Figure I.1.

Box I.1. Burkina Faso—Macroeconomic Assumptions

Real GDP growth is projected to average 6 percent in the medium and long run, broadly in line with the historical average of the past ten years. In 2008, surging oil and food prices will keep growth below average at about 4.5 percent, while increasing annual average **inflation** to above 6 percent. In the following years, inflation is expected to return to its historical average of about 2 percent.

The **overall fiscal deficit**, with and without grants, is projected to decrease over time as revenue increases towards the WAEMU tax revenue target of 17 percent of GDP. The revenue increase is supported by ongoing efforts to strengthen tax and customs administration, which is a centerpiece of the current Fund-supported program, and comprehensive tax policy reforms planned for coming years. Expenditures have increased significantly since 2006 and are projected to remain high to meet the authorities' PRSP objectives. External financing requirements are to be met equally by grants and loans, which is a conservative assumption based on recent experience. The decline in the overall deficit including grants translates into a broadly equal reduction in external new borrowing, because domestic financing is assumed to be zero for most years as its costs are high.



The **current account deficit** is expected to narrow moderately in the medium and long run as the tighter fiscal stance dampens import demand. In the short run, higher oil prices lead to a significant increase in imports and widening of the current account deficit; this is expected to be partially reversed once the connection of the transmission grid with Côte d'Ivoire is completed, which should reduce the consumption of oil for electricity generation. The exports-to-GDP ratio will decline significantly in 2008 because of a poor cotton harvest in 2007; in the medium term, gold exports due to the opening of new commercial mining operations will support export growth.

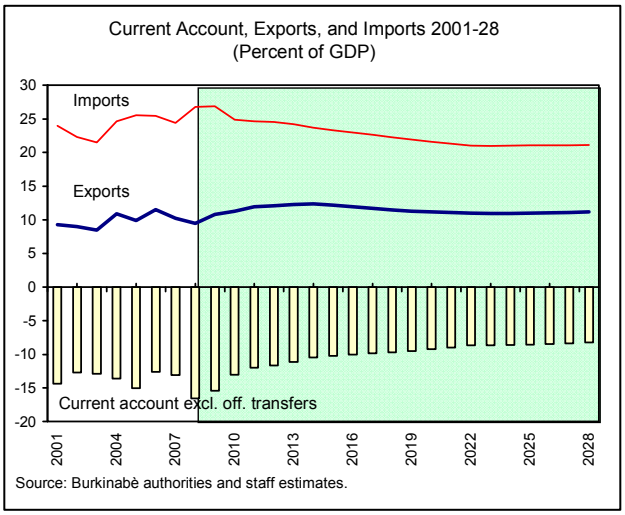
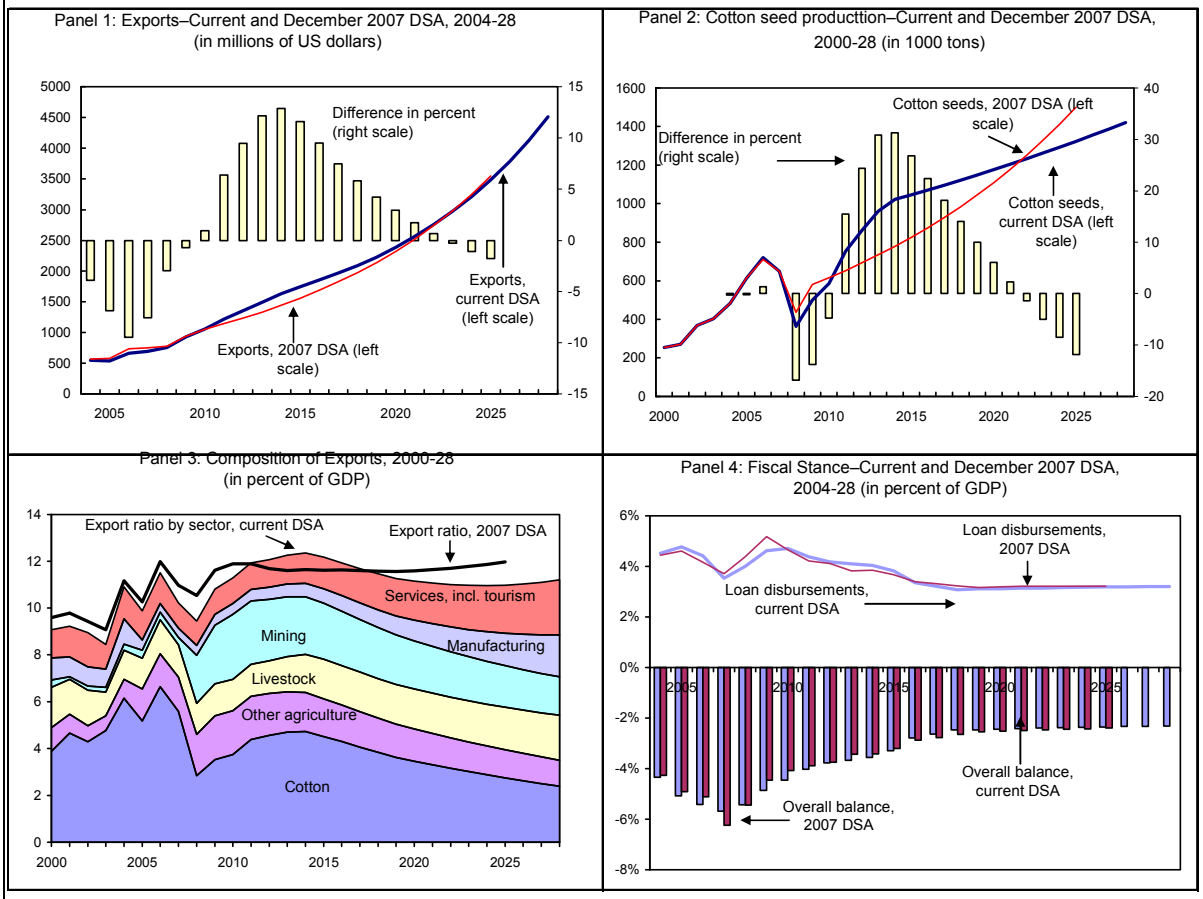


Figure I.1. Burkina Faso: Changes in Macroeconomic Framework Between Current and Previous DSA



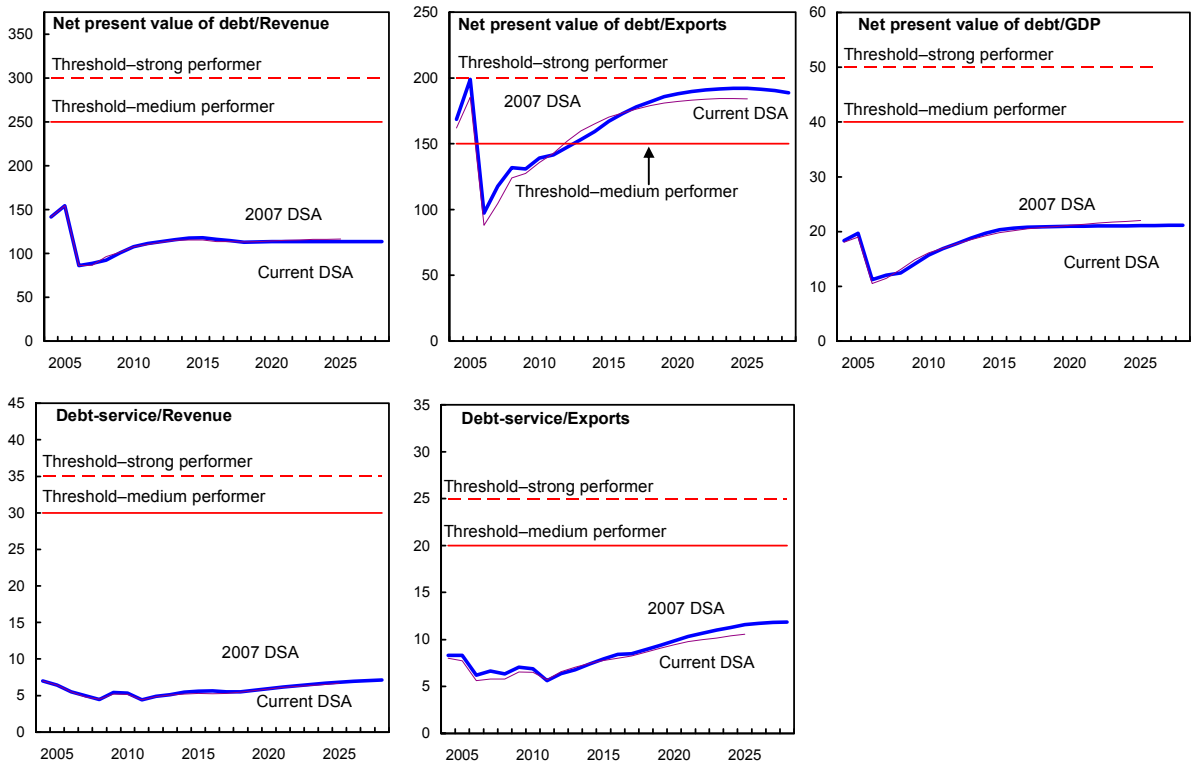
Sources: Burkinabè authorities; and IMF staff estimates.

4. **The fiscal path is broadly unchanged**. Externally-financed capital expenditures in 2007 remained below expectations, leading to a modest downward revision in the following years and correspondingly lower borrowing requirements. Fiscal targets were otherwise maintained, with a long-run fiscal deficit target of about 2.5 percent of GDP (Figure I.1, Panel 4).

5. **The debt database has been updated and reconciled with creditor statements, but this has not led to a major revision in the NPV of debt**. The debt database has been updated to reflect the debt stock outstanding as of end-2007; the reconciliation with available creditor statements was done jointly by staffs of the African Development Bank, the Fund, and the World Bank. The resulting change in the estimated NPV of debt is small—for 2007, it has been revised upwards by about 3.75 percent, whereas the average revision for 2007–25 is approximately -0.5 percent.

6. **The lower DSF thresholds signal a higher risk of debt distress (Figure I.2)**. With few changes in the macroeconomic framework, the path for most DSF indicators is close to previous projections. The country's risk of debt distress, however, has increased because all DSF indicators are now closer to their respective thresholds than was previously the case and the NPV of debt-to-exports ratio even breaches its threshold.

Figure I.2. Burkina Faso: Selected Debt Indicators (Baseline Scenario)–Current and December 2007 DSA, 2004–28 (percent)



Source: Burkinabè authorities and IMF–World Bank staff estimates.

B. Assessing Vulnerabilities

7. **This section explores the vulnerability of the debt sustainability outlook to lower export and growth assumptions and maintaining the current fiscal stance**.

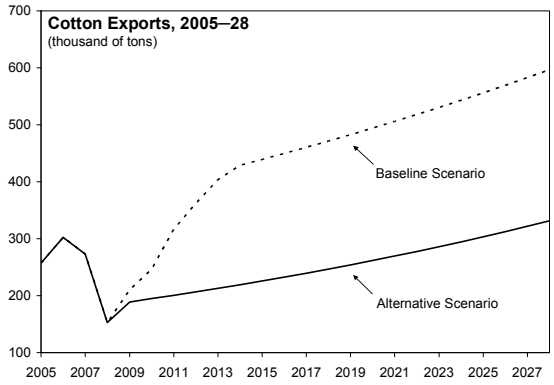
8. **A worse-than-expected export and growth performance is a clear risk, but simulations show that most DSF indicators are relatively robust to more unfavorable export and growth scenarios, with the exception of the NPV of debt-to-exports ratio**. The baseline scenario of the DSA assumes (i) a rapid cotton production rebound to pre-crisis levels, and (ii) much higher gold prices in line with WEO projections. The country-specific alternative scenario in Figure I.3 relaxes both assumptions and lowers the GDP growth rate, while maintaining the path for the overall fiscal deficit in percent of GDP. The impact on most debt sustainability indicators is modest, with baseline projections remaining below their indicative thresholds for medium performers (Figure I.4). The exception is the NPV of debt-to-exports ratio, which breaches the threshold.

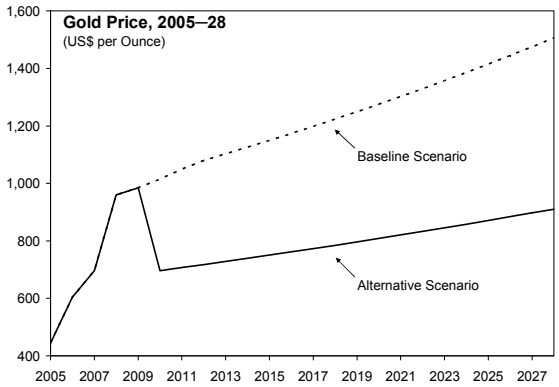
9. **A key factor for the resilience of most debt indicators to the adverse export and growth shock is the assumption that the fiscal deficit declines throughout the projection period as programmed in the baseline**. This implicitly assumes that (i) structural reforms in the area of tax administration and fiscal policy would be implemented and (ii) that fiscal tightening would be adopted as a response to permanent losses of revenue.

10. **If the fiscal deficit remains at the level observed over the past few years, debt sustainability will be at risk**. In a second country-specific scenario, growth and exports follow the same trend as in the baseline, but fiscal policy is less tight, with the overall fiscal deficit-to-GDP ratio remaining in the long-run at about 4.5 percent of GDP, vs. 2.5 percent of GDP in the baseline (Figure I.5). This scenario leads to an upward trend in all debt indicators, and a number of indicative thresholds would be breached. Besides increasing the risk of debt distress, the economy would become much more vulnerable to negative external shocks, and there would be little room for implementing counter-cyclical policies.

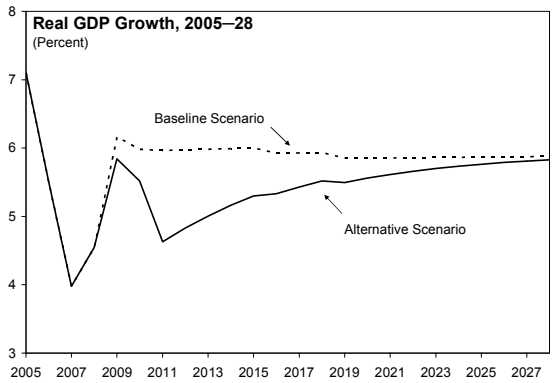
Figure I.3. Alternative Scenario with Lower Exports and Growth: Assumptions

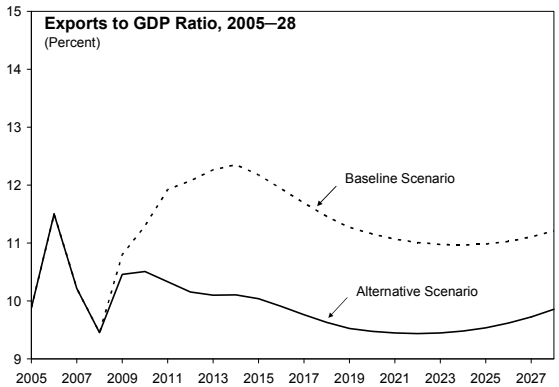
Cotton production and gold prices are lower over the medium and long term, ...



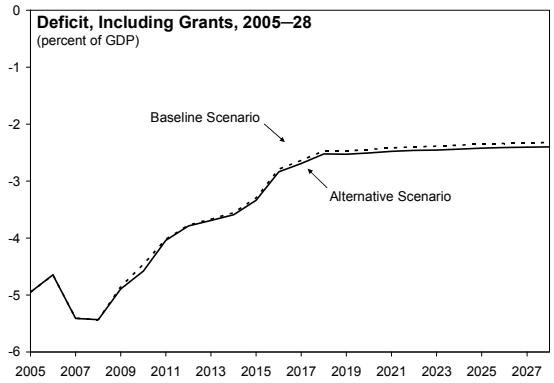


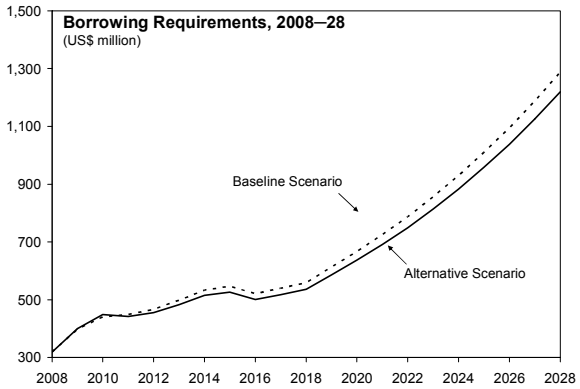
… leading to lower growth and a reduced export-to-GDP ratio.





The fiscal deficit, in percent of GDP, follows the baseline, but nominal borrowing requirements are reduced because of the lower GDP level.

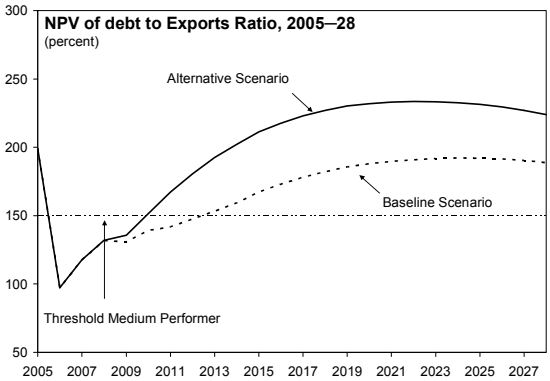




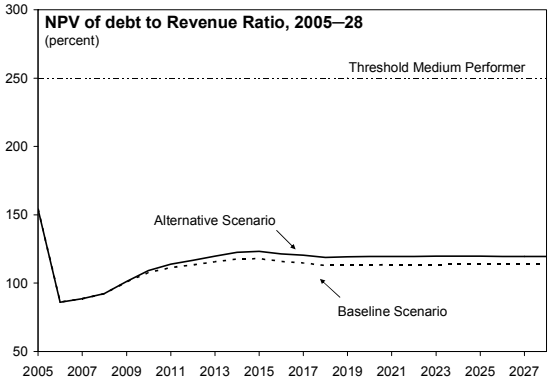
Sources: Authorities, World Bank and IMF staff estimates.

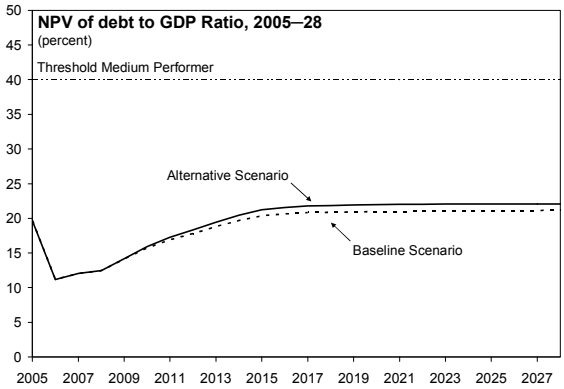
Figure I.4. Alternative Scenario with Lower Exports and Growth: Results

Because of a less favorable outlook for exports, the NPV of debt-to-exports ratio would exceed 200 percent, the threshold for strong performers …

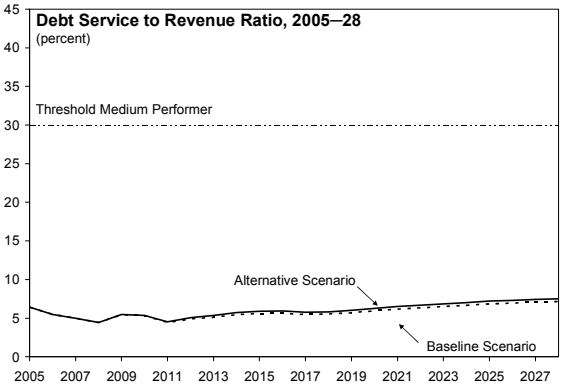


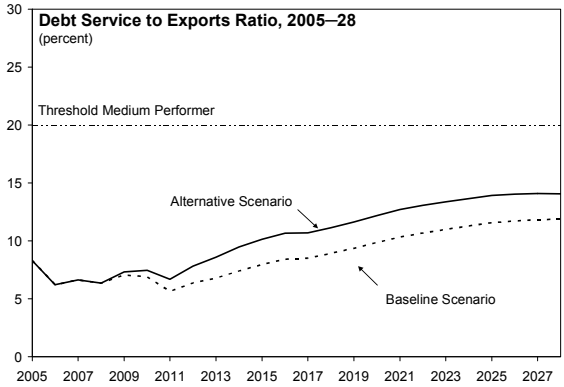
… but debt would remain otherwise sustainable …





… as Burkina Faso would preserve its capacity to meet its external debt service payments.

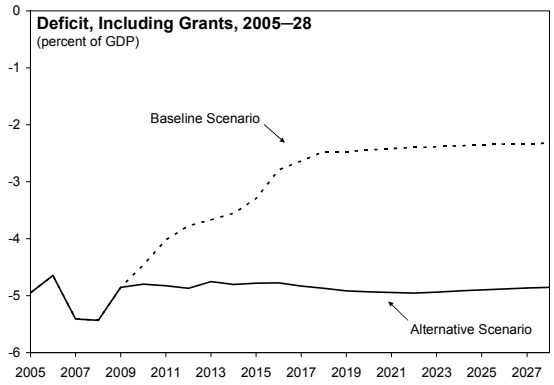


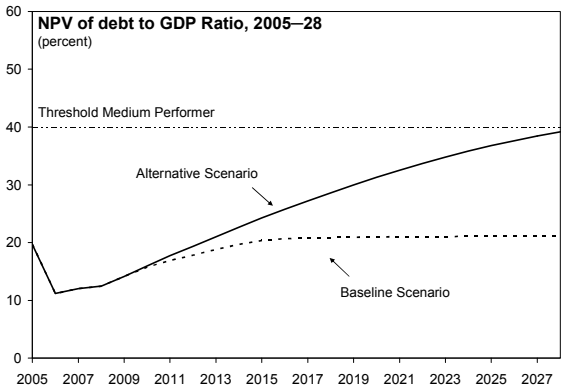


Sources: Authorities, World Bank and IMF staff estimates.

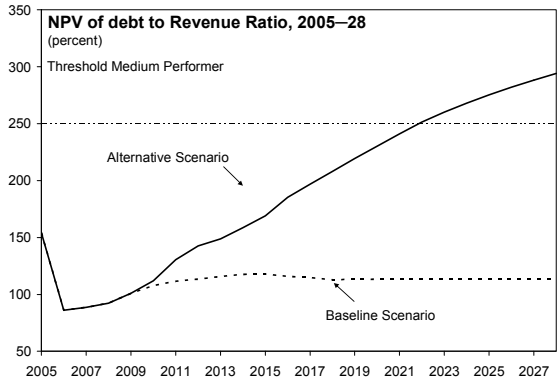
Figure I.5. Alternative Scenario with Higher Fiscal Deficit

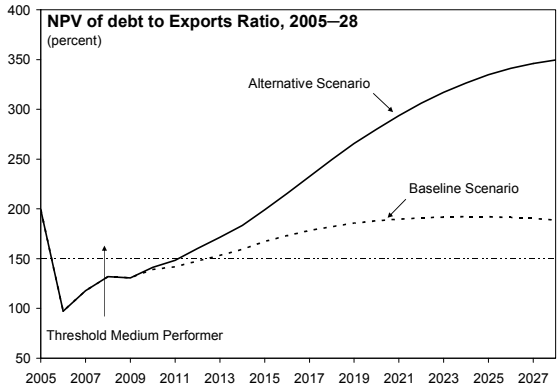
As a result of a higher fiscal deficit the NPV of debt to GDP ratio will increase continuously...



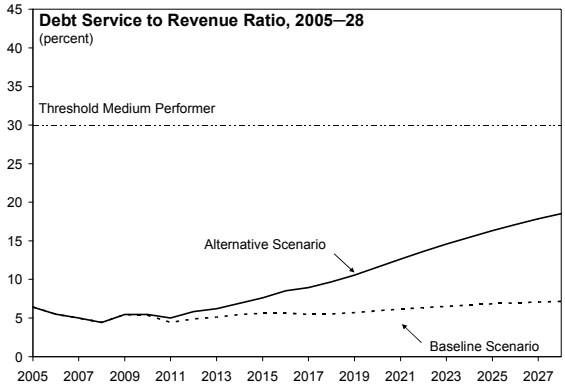


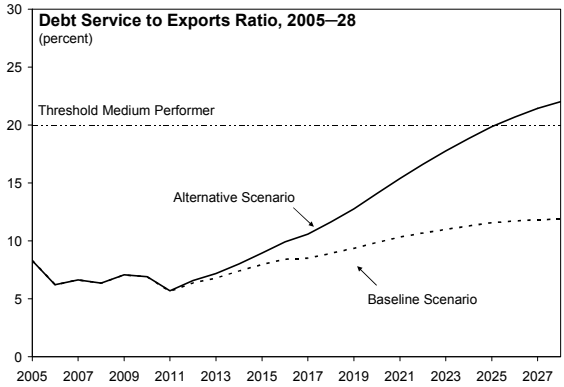
... and debt sustainability will be at risk, as underlined with the two other NPV of debt ratios, which will breach their indicative thresholds.





Debt service ratios will also increase continuously.





Sources: Authorities, World Bank and IMF staff estimates.

C. Risk of Debt Distress

11. **The risk of debt distress is high because one indicative threshold is breached in the baseline projection: the threshold for the NPV of debt-to-exports ratio**. The peak ratio of about 190 percent in the baseline projection of the NPV of debt-to-exports ratio is well above the 150 percent threshold. Under DSF guidelines, such a breach warrants a classification of high risk of debt distress, although the GDP- and revenue-based DSF indicators remain well below their indicative thresholds, including some of the alternative scenarios and stress tests.

12. **There are a number of other positive factors as well as some downside risks:**

Positive factors

- Outstanding payment record: since the transition from a centralized to a market-oriented economy in the early 1990s, Burkina Faso's government has consistently demonstrated its commitment to stay current on its external payment obligations. This includes the period before the country benefited from debt relief under the Heavily Indebted Poor Country Initiative (HIPC), when the NPV of debt-to-exports ratio was well above 200 percent. Despite a high debt burden, the government even reduced outstanding external payment arrears in these years. Given this demonstrated payment record, it appears unlikely that the government will fail to service its debt obligations. Moreover, the debt service ratios are projected to remain low throughout the entire projection period.
- Additional grant financing: this could improve the debt sustainability outlook significantly. The share of grant financing excluding IDA and AFD has increased in recent years to about 55 percent, whereas DSA projections are based on a more conservative 50 percent grant-financing share. Raising the grant-financing share to 55 percent in the projection period would lower the peak NPV of debt-to-exports ratio by about 20 percentage points, but it would still remain above the indicative threshold.
- Improvements in debt management: the authorities have recently adopted a national debt policy and are currently receiving technical assistance in this area from the World Bank. Preliminary conclusions from the Debt Management Performance Assessment (DEMPA) indicate that Burkina Faso has relatively strong controls for the contracting of external debt. The 35 percent concessionality floor is strictly enforced for all central-government foreign-currency borrowings, and projects are consistently scrutinized by a National Public Debt Committee before negotiations are concluded. A debt strategy that is planned to be added as an annex to the 2009 budget is being finalized and will include a ceiling on new borrowings based on the analysis of debt sustainability. While the evaluation also identifies a number of areas that do not meet the minimum requirements set out in the DeMPA framework (overstretching

of the Public Debt National Committee, poor information on disbursements, poor dissemination on debt data, and weak audit institutions with respect to external borrowing), in many cases the improvements required for reaching those standards are relatively modest.

Downside risks

- Delayed fiscal adjustment: this poses a significant downside risk, because without deficit reduction Burkina Faso's debt burden could eventually become unsustainable, even measured relative to GDP or revenue. Preventing this will require a stronger revenue effort, which in turn depends on further efforts in strengthening tax and customs administration, and on implementing substantial tax policy reform.
- Adverse export shocks: these shocks would mostly affect the export-based debt indicators, whereas the other indicators are relatively robust with respect to this type of shock. Given that Burkina Faso is unlikely to have problems mobilizing foreign exchange to service its public debt[5], the impact of this type of shock on the risk of debt distress is limited.

13. **In sum, there are significant up- and downside-risks**. The most significant downside risk is delayed fiscal adjustment, in which case the NPV of debt-to-exports ratio could climb above 300 percent (Figure I.5). However, even under this dire scenario a debt default may not be the most likely outcome because (i) additional grants, particularly those related to the IDA/AFD grant allocation mechanism, would mitigate the increase in debt ratios, and (ii) the authorities have proven their commitment in the past to stay current on their debt allocations even when the debt burden is high.

II. FISCAL DEBT SUSTAINABILITY ANALYSIS

14. **The results for the fiscal DSA are broadly similar to those of the external DSA**. Failure to reduce the current fiscal deficits would lead to ever-increasing debt indicators, which is clearly unsustainable. The domestic debt stock is low—gross domestic debt is estimated at approximately 3.5 percent of GDP at end-2007, and net debt at 0.5 percent of GDP; consequently, the baseline scenarios for external and total public debt are similar. The alternative scenarios in the fiscal DSA, though, differ in exploring the impact of different fiscal policies on the debt sustainability outlook. For example, the 'no-reform' scenario, which maintains the 2008 primary deficit of 5 percent of GDP over the projection horizon, reinforces the result above (Figure I.7).

[5] Burkina Faso's membership in the West African Economic and Monetary Union (WAEMU) gives it access to the union's pool of foreign exchange reserves.

III. CONCLUSION

15. **Burkina Faso's risk of debt distress is high** on account of the NPV of debt-to-exports ratio; all other debt indicators are comfortably below their policy-dependent thresholds under the baseline scenario. The overall deterioration from the previous DSA suggests that more emphasis needs to be given to a prudent fiscal policy, to external borrowing limited to concessional loans, and to a continued build-up in public debt management capacity.

Appendix I—Table 1. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2008–28 1/

(percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008–13 Average	2018	2028	2014–28 Average
External debt (nominal) 1/	**38.9**	**21.1**	**19.7**			**20.7**	**23.5**	**26.1**	**28.0**	**29.3**	**30.6**		**32.7**	**31.9**	
o/w public and publicly guaranteed (PPG)	38.9	21.1	19.7			20.7	23.5	26.1	28.0	29.3	30.6		32.7	31.9	
Change in external debt	-2.9	-17.8	-1.3			1.0	2.8	2.6	1.9	1.3	1.3		-0.1	-0.1	
Identified net debt–creating flows	8.0	6.7	0.4			8.0	9.3	8.5	7.3	6.8	5.6		3.2	2.9	
Non–interest current account deficit	**11.4**	**9.3**	**8.9**	**10.1**	**1.1**	**12.7**	**12.7**	**10.4**	**9.4**	**9.1**	**8.6**		**7.6**	**6.4**	7.1
Deficit in balance of goods and services	15.6	14.0	14.2			17.4	16.1	13.6	12.7	12.4	12.0		10.8	9.9	
Exports	9.9	11.5	10.2			9.5	10.8	11.3	11.9	12.1	12.3		11.5	11.2	
Imports	25.5	25.5	24.4			26.8	26.9	24.9	24.7	24.5	24.2		22.3	21.1	
Net current transfers (negative = inflow)	-4.3	-4.4	-5.3	-4.7	0.8	-4.7	-3.4	-3.1	-3.3	-3.3	-3.3		-3.1	-3.5	-3.3
o/w official	-3.4	-3.0	-4.0			-3.7	-2.3	-2.2	-2.4	-2.2	-2.2		-1.6	-1.5	
Other current account flows (negative = net inflow)	0.1	-0.2	0.0			0.0	0.0	0.0	0.0	-0.1	-0.1		-0.1	0.0	
Net FDI (negative = inflow)	**-0.6**	**-0.6**	**-5.5**	**-1.0**	**1.6**	**-4.0**	**-2.6**	**-1.0**	**-0.9**	**-1.0**	**-1.7**		**-3.1**	**-2.2**	-2.4
Endogenous debt dynamics 2/	**-2.8**	**-2.0**	**-2.9**			**-0.6**	**-0.8**	**-0.9**	**-1.1**	**-1.2**	**-1.3**		**-1.4**	**-1.3**	
Contribution from nominal interest rate	0.3	0.2	0.2			0.2	0.4	0.4	0.3	0.3	0.3		0.4	0.5	
Contribution from real GDP growth	-2.8	-2.0	-0.7			-0.8	-1.2	-1.3	-1.4	-1.5	-1.6		-1.8	-1.7	
Contribution from price and exchange rate changes	-0.3	-0.3	-2.4			…	…	…	…	…	…		…	…	
Residual (3–4) 3/	**-10.9**	**-24.6**	**-1.7**			**-7.0**	**-6.5**	**-6.0**	**-5.5**	**-5.5**	**-4.2**		**-3.2**	**-3.0**	
o/w exceptional financing	0.0	-23.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	19.7	11.2	12.0			12.5	14.1	15.7	16.9	17.8	18.8		20.9	21.2	
percent of exports	199.1	97.3	117.7			131.8	130.8	139.2	141.8	147.5	153.3		182.1	188.8	
NPV of PPG external debt	**19.7**	**11.2**	**12.0**			**12.5**	**14.1**	**15.7**	**16.9**	**17.8**	**18.8**		**20.9**	**21.2**	
percent of exports	**199.1**	**97.3**	**117.7**			**131.8**	**130.8**	**139.2**	**141.8**	**147.5**	**153.3**		**182.1**	**188.8**	
percent of government revenues	**154.2**	**86.1**	**88.6**			**92.3**	**100.9**	**107.7**	**111.4**	**113.3**	**115.7**		**112.9**	**113.7**	
Debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.6**			**6.3**	**7.1**	**6.9**	**5.6**	**6.4**	**6.8**		**8.9**	**11.9**	
PPG debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.6**			**6.3**	**7.1**	**6.9**	**5.6**	**6.4**	**6.8**		**8.9**	**11.9**	
PPG debt service-to-revenue ratio (percent)	**6.4**	**5.5**	**5.0**			**4.4**	**5.4**	**5.3**	**4.4**	**4.9**	**5.1**		**5.5**	**7.2**	
Total gross financing need (US$ billions)	631.1	548.6	271.1			735.8	938.5	957.9	937.6	988.2	933.1		1014.0	2218.3	
Non–interest current account deficit that stabilizes debt ratio	14.3	27.2	10.2			11.7	9.9	7.9	7.5	7.7	7.3		7.7	6.5	
Key macroeconomic assumptions															
Real GDP growth (percent)	7.1	5.5	4.0	5.6	1.9	4.5	6.2	6.0	6.0	6.0	6.0	5.8	5.9	5.9	5.9
GDP deflator in US$ terms (percent change)	0.8	0.7	12.8	5.3	10.3	12.7	1.8	2.6	3.1	3.2	2.3	4.3	2.3	2.2	2.3
Effective interest rate (percent) 5/	0.7	0.7	1.2	0.6	0.4	1.1	1.8	1.8	1.2	1.2	1.3	1.4	1.4	1.5	1.4
Growth of exports of G&S (US$ terms, percent)	-2.1	23.7	4.2	12.0	22.3	8.9	23.5	13.7	15.3	10.7	10.2	13.7	6.3	9.3	7.7
Growth of imports of G&S (US$ terms, percent)	11.7	6.1	12.3	11.0	12.8	29.5	8.3	0.6	8.4	8.7	7.2	10.4	6.7	8.4	7.3
Grant element of new public sector borrowing (percent)	…	…	…	…	…	32.4	33.2	32.5	32.3	31.9	31.4	32.3	29.9	27.7	29.2
Aid flows (US$ billions) 7/	554.8	564.6	678.6			835.9	891.5	965.1	997.3	1033.8	1046.1		1117.8	2574.3	
o/w Grants	249.8	280.2	439.4			524.3	496.9	526.0	548.7	566.9	548.1		558.9	1287.1	
o/w Concessional loans	259.4	254.9	239.2			311.6	394.6	439.1	448.7	466.9	498.1		558.9	1287.1	
Grant–equivalent financing (percent of GDP) 8/	…	…	…			7.9	7.3	7.1	6.8	6.4	5.8		4.0	4.1	4.2
Grant–equivalent financing (percent of external financing) 8/	…	…	…			73.6	70.2	69.2	69.5	69.2	67.4		65.0	63.8	64.6
Memorandum items:															
Nominal GDP (US$ billions)	5436.9	5776.6	6776.8			7981.4	8625.6	9375.6	10245.5	11203.4	12151.1		18238.9	40248.1	
(NPVt–NPVt–1)/GDPt–1 (percent)						2.6	2.8	3.0	2.7	2.6	2.6	2.7	1.8	1.8	1.9

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r – g – r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections, capital transfers are included–in particular project grants, which are projected to average about 2.5 percent of GDP over the long term, and about 4 percent over 2008-12 due to MCC grants—and private, non-debt-creating capital inflows. Projections also includes contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current–year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant–equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Appendix I. Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	**2018**	2028
NPV of debt-to-GDP ratio								
Baseline	12	14	16	17	18	19	**21**	21
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	12	13	14	15	16	18	**29**	40
A2. New public sector loans on less favorable terms in 2008–27 2/	12	15	18	20	22	24	**29**	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	12	14	16	18	19	20	**22**	22
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	12	16	21	22	22	23	**24**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	12	15	18	20	21	22	**24**	24
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	12	16	18	19	20	21	**22**	21
B5. Combination of B1–B4 using one–half standard deviation shocks	12	15	19	20	21	22	**24**	23
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	12	20	22	24	25	27	**30**	30
NPV of debt-to-exports ratio								
Baseline	132	131	139	142	148	153	**182**	189
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	132	120	123	127	137	151	**257**	360
A2. New public sector loans on less favorable terms in 2007–26 2/	132	142	161	171	184	197	**256**	297
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	132	131	139	142	148	153	**182**	189
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	132	205	322	319	325	331	**358**	337
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	132	131	139	142	148	153	**182**	189
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	132	148	160	160	165	170	**192**	191
B5. Combination of B1–B4 using one–half standard deviation shocks	132	169	219	220	227	233	**265**	263
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	132	131	139	142	148	153	**182**	189
NPV of debt-to-revenue ratio								
Baseline	92	101	108	111	113	116	**113**	114
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	92	92	95	100	105	114	**159**	217
A2. New public sector loans on less favorable terms in 2007–26 2/	92	110	124	135	142	149	**159**	179
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	92	103	113	116	118	121	**118**	119
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	92	115	143	144	143	143	**127**	117
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	92	108	125	129	131	134	**131**	132
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	92	114	124	126	127	128	**119**	115
B5. Combination of B1–B4 using one–half standard deviation shocks	92	110	131	134	135	136	**127**	123
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	92	143	153	158	161	164	**160**	161

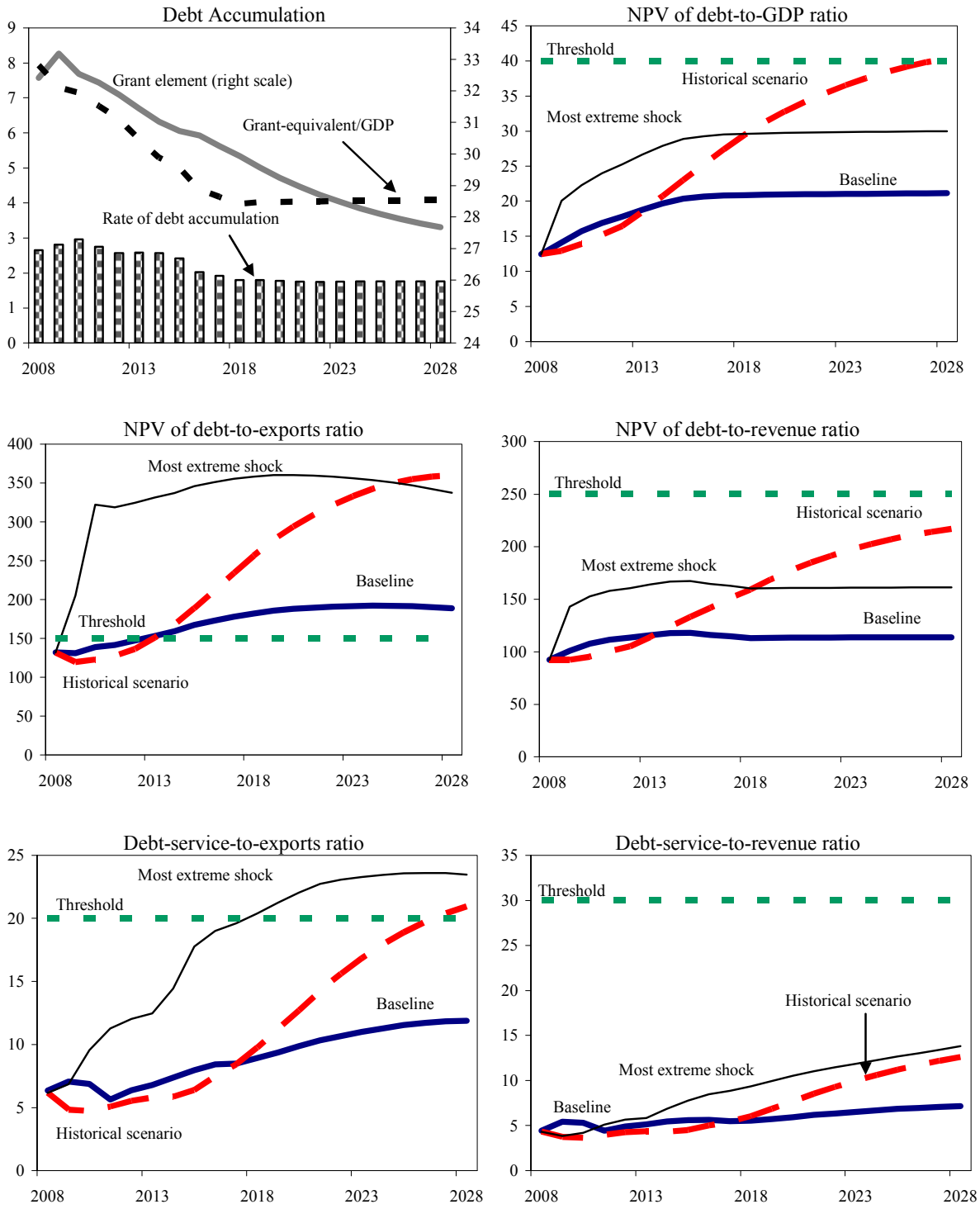
Appendix I. Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28 (concluded)
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
Debt service-to-exports ratio								
Baseline	6	7	7	6	6	7	**9**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	6	5	5	5	6	6	**10**	21
A2. New public sector loans on less favorable terms in 2008–27 2/	6	5	5	6	7	8	**15**	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	6	5	5	6	6	6	**9**	12
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	6	7	10	11	12	12	**20**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	6	5	5	6	6	6	**9**	12
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	6	5	5	6	6	7	**10**	13
B5. Combination of B1–B4 using one–half standard deviation shocks	6	6	7	8	9	9	**14**	18
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	6	5	5	6	6	6	**9**	12
Debt service-to-revenue ratio								
Baseline	4	5	5	4	5	5	**6**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	4	4	4	4	4	4	**6**	13
A2. New public sector loans on less favorable terms in 2008–27 2/	4	4	4	5	6	6	**9**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	4	4	4	5	5	5	**6**	8
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	4	4	4	5	5	5	**7**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	4	4	5	5	5	6	**7**	9
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	4	4	4	5	5	5	**6**	8
B5. Combination of B1–B4 using one–half standard deviation shocks	4	4	4	5	5	5	**7**	8
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	4	5	6	6	7	7	**8**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	**33**	33

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (US$ terms), non–interest current account in percent of GDP, and non–debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumingan offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure I.6. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios (in percent), 2008-28 1/



Source: Staff projections and simulations.
1/ Thresholds for medium performer.

Appendix I–Table 3.Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-28

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2005	2006	2007	Historical Average 5/	Standard Deviation 5/	2008	2009	2010	2011	2012	2013	2008-13 Average	2018	2028	2014-28 Average
Public sector debt 1/	44.4	22.8	22.0			24.0	26.4	28.7	30.3	31.4	32.5		34.0	32.5	
o/w foreign-currency denominated	40.9	20.0	18.6			20.7	23.5	26.0	27.8	29.1	30.5		32.5	31.7	
Change in public sector debt	0.3	-21.6	-0.8			2.0	2.4	2.3	1.6	1.1	1.2		-0.2	-0.2	
Identified debt-creating flows	6.6	-23.0	2.3			3.3	3.0	2.3	1.6	1.2	1.2		-0.2	-0.2	
Primary deficit	4.6	5.0	5.3	3.6	1.2	5.0	4.3	3.9	3.6	3.3	3.2	3.9	2.0	1.8	2.1
Revenue and grants	17.4	18.9	20.1			20.1	19.8	20.2	20.5	20.8	20.8		21.5	21.8	
of which: grants	4.6	5.9	6.5			6.6	5.8	5.6	5.4	5.1	4.5		3.1	3.2	
Primary (noninterest) expenditure	21.9	23.9	25.3			25.0	24.1	24.1	24.1	24.1	24.0		23.5	23.6	
Automatic debt dynamics	2.1	-5.9	-2.9			-1.7	-1.3	-1.6	-2.0	-2.1	-2.0		-2.1	-2.0	
Contribution from interest rate/growth differential	-3.6	-3.1	-1.1			-1.0	-1.3	-1.4	-1.7	-1.8	-1.9		-2.0	-1.9	
of which: contribution from average real interest rate	-0.7	-0.8	-0.2			0.0	0.1	0.1	-0.1	-0.1	-0.1		-0.1	-0.1	
of which: contribution from real GDP growth	-2.9	-2.3	-0.9			-1.0	-1.4	-1.5	-1.6	-1.7	-1.8		-1.9	-1.8	
Contribution from real exchange rate depreciation	5.7	-2.8	-1.9			-0.7	0.0	-0.2	-0.3	-0.3	-0.1		...	...	
Other identified debt-creating flows	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-6.3	1.4	-3.1			-1.3	-0.6	-0.1	0.0	-0.1	-0.1		0.0	0.0	
NPV of public sector debt	23.2	14.0	15.5			15.8	17.1	18.5	19.4	20.0	20.9		22.3	21.9	
o/w foreign-currency denominated	19.7	11.2	12.0			12.5	14.1	15.7	16.9	17.8	18.8		20.9	21.2	
o/w external	19.7	11.2	12.0			12.5	14.1	15.7	16.9	17.8	18.8		20.9	21.2	
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 2/	6.0	6.2	6.6			5.8	5.3	4.8	4.4	4.2	4.2		3.1	3.2	
NPV of public sector debt-to-revenue and grants ratio (in percent)	133.6	73.9	77.1			78.5	86.6	91.3	94.4	96.5	100.6		103.6	100.5	
NPV of public sector debt-to-revenue ratio (in percent)	181.7	107.3	113.9			116.7	122.2	126.4	127.7	127.6	128.5		120.8	117.8	
o/w external 3/	154.2	86.1	88.6			92.3	100.9	107.7	111.4	113.3	115.7		112.9	113.7	
Debt service-to-revenue and grants ratio (in percent) 4/	8.2	6.2	6.7			4.4	4.7	4.6	4.0	4.3	4.6		5.1	6.3	
Debt service-to-revenue ratio (in percent) 4/	11.2	8.9	9.9			6.6	6.6	6.4	5.3	5.7	5.8		6.0	7.3	
Primary deficit that stabilizes the debt-to-GDP ratio	4.3	26.6	6.1			2.9	1.9	1.6	2.0	2.2	2.0		2.2	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.1	5.5	4.0	5.6	1.9	4.5	6.2	6.0	6.0	6.0	6.0	5.8	5.9	5.9	5.9
Average nominal interest rate on forex debt (in percent)	0.7	0.6	1.1	0.6	0.4	1.1	1.8	1.8	1.2	1.2	1.3	1.4	1.4	1.5	1.4
Average real interest rate on domestic currency debt (in percent)	3.9	5.1	4.1	-0.3	5.2	4.1	4.1	3.6	3.3	3.5	3.3	3.6	3.1	2.6	2.9
Real exchange rate depreciation (in percent, + indicates depreciation)	16.0	-7.4	-10.0	-2.6	12.5	-3.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.6	-0.1	3.4	2.8	2.9	5.0	1.3	1.8	2.1	2.1	2.3	2.4	2.3	2.2	2.3
Growth of real primary spending (deflated by GDP deflator, in percent)	10.8	15.0	10.2	7.4	6.6	3.3	2.2	6.1	6.0	5.9	5.4	4.8	6.1	5.9	5.8
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	0.0	0.0	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

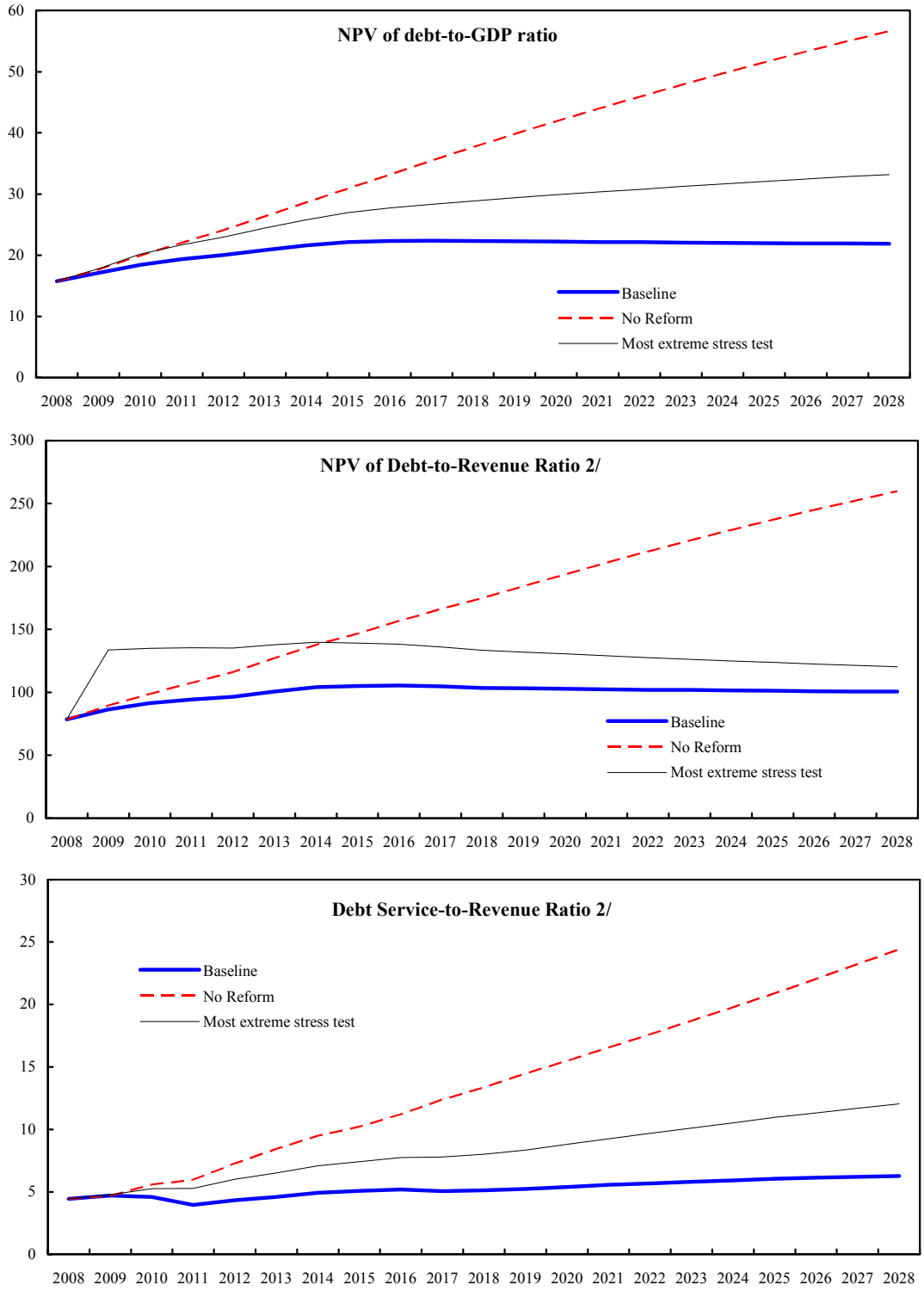
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Appendix I–Table 4.Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt 2008-28

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
NPV of Debt-to-GDP Ratio								
Baseline	16	17	18	19	20	21	22	22
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	16	18	19	20	21	27	39
A2. Primary balance is unchanged from 2008	16	18	20	22	24	26	38	57
A3. Permanently lower GDP growth 1/	16	17	19	20	21	22	26	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	16	18	20	22	23	24	29	33
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	16	18	20	21	21	22	23	23
B3. Combination of B1-B2 using one half standard deviation shocks	16	17	19	20	20	21	22	21
B4. One-time 30 percent real depreciation in 2009	16	22	22	22	22	22	22	21
B5. 10 percent of GDP increase in other debt-creating flows in 2009	16	26	27	28	28	29	29	26
NPV of Debt-to-Revenue Ratio 2/								
Baseline	78	87	91	94	96	101	104	100
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	78	83	87	90	94	100	127	178
A2. Primary balance is unchanged from 2008	78	89	99	107	116	127	175	260
A3. Permanently lower GDP growth 1/	78	87	93	97	100	106	122	161
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	78	89	99	105	109	117	133	151
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	78	89	97	100	102	106	108	103
B3. Combination of B1-B2 using one half standard deviation shocks	78	86	93	95	96	100	101	94
B4. One-time 30 percent real depreciation in 2009	78	109	109	107	106	107	104	98
B5. 10 percent of GDP increase in other debt-creating flows in 2009	78	134	135	135	135	138	134	120
Debt Service-to-Revenue Ratio 2/								
Baseline	4	5	5	4	4	5	5	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	5	3	3	4	5	8	14
A2. Primary balance is unchanged from 2008	4	5	6	6	7	8	13	24
A3. Permanently lower GDP growth 1/	4	5	5	4	5	5	7	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	4	5	5	5	6	7	8	12
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	4	5	5	6	5	5	5	7
B3. Combination of B1-B2 using one half standard deviation shocks	4	5	4	4	4	4	5	5
B4. One-time 30 percent real depreciation in 2009	4	5	6	5	5	6	7	8
B5. 10 percent of GDP increase in other debt-creating flows in 2009	4	5	21	10	8	7	7	9

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure I.7.Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2008-28 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2018.
2/ Revenue including grants.